



Final Presentation

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LOOKING FOR EVIDENCE

It is important to provide evidence showing P.S. has really affected the lives of the people (firm's employees)

For microfinance business, we want to answer: whether the PS is a cost effective way to have better client's credit or not

REPORTS

Model report for the firm's quantitative data analysis

Cost estimation

Impact evaluation of Poverty Stoplight (PS) on microfinance clients

Key findings from literature about the work environment and firm's productivity

- What makes a good job? Earning quality, Labor market security, Quality of working environment, in below picture you can compare the **OECD job quality framework** (Sandrine Cazes, 2015) vs **Poverty Stoplight**
- A positive relation between the average **employee's trust** to their manager and the three measure of **work performance** ; financial performance, labor productivity and product/service quality (S. Brown et al 2015)
- Measure trust indirectly by these indicators: Staff communication, profit-sharing, internal promotion, staff turnover and training. (M. Goergen et al, 2012)
- A company can **improve its productivity** by restructuring toward a **healthier work environment** (J. Ruizac et al, 2015)

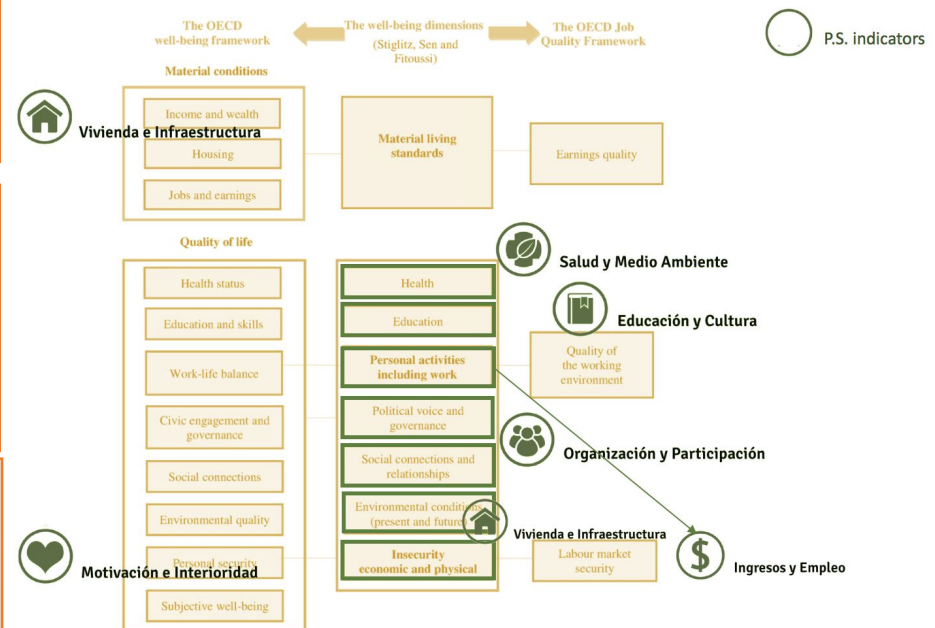
Based on these papers and in order to show the impact of the poverty stoplight, we create a "Model Report" with below indicators

Firms' indicators:
Productivity
profit

Employees' indicators:
Staff turn-over
Education
Seniority
Absence, arrival delay
Average Poverty score

From qualitative analysis (with Emily) we heard about the impact of PS on

- Increase firm's brand value
- A new channel for marketing
- Expand markets => which was obviously one of main reason for CAFEPAR!



Source: The OECD job quality framework (Sandrine Cazes, 2015)

Model report for firm's data

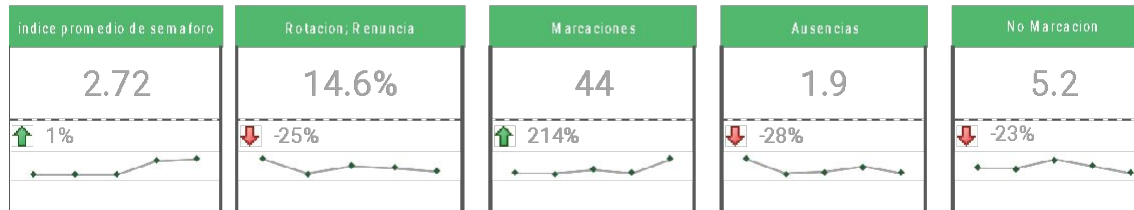
Reporte Annual

CafePar

la primera encuesta de
semáforo fue en 2016

2017

llaves Metricas



Todas las Métricas

Métrico	LASTYEAR (2017)	LASTYEAR (2016)	% Cambio	Tendencia de 5 Años
Productividad (\$/ Persona)	0.0	0.0		
Rotacion; Renuncia	14.6%	19.5%	↓ -25%	
Marcaciones	44.0	14.0	↑ 214%	
Ausencias	1.9	2.6	↓ -28%	
No Marcacion	5.2	6.8	↓ -23%	
índice promedio de sema	2.7	2.7	↑ 1%	
Deudas personales prome	0.0	0.0		
Ahorros personal promedi	0.0	0.0		
Promedio de enfermedad	0.0	0.0		
Beneficio	0.0	0.0		
Employees Seniority	0.0	0.0		

This is a model of analysis that can replicates to all firms Angel is now doing for 2nd firm

Trends are good for 4 indicators
Not attribution of causality

Also this can be done in other countries with more historical data (Chile, ...)

This Model Report contains the result of few data that we received from CAFEPAR

For future engagement with businesses, we suggest to motivate them to keep track of these indicators

- Businesses seems satisfied with team approach and sharing data till now, for more motivation
- Also PS team can hold annual competitions among firms to nominate the best PS implementation/improvement => you can annually ask for the interested indicators

Based on 2017 data

COST ESTIMATION: Case based analysis approach

		Based on 2017 data
Configuration	To compare the range Please Choose a value for both lists	
Dedication of asesora's time to semaforo only - (First Number is based on the asesora questionnaire's result)		25%
Dedication of Headquarter cost for the semaforo only- (First Number is the ratio of "sum of the salaries of semaforo team excluding replicas/total HQ salaries)		3%
Output	Cost Amount	
Estimated competitions Cost - For 1 year - But it has been financed by sponsor since now	Gs.	(201,000,000)
Forum, meetings and Training Cost - For 1 year	Gs.	(67,684,550)
Regional Offices cost share dedicated to the semaforo - For 2017	Gs.	(5,366,212,916)
Head Quarter dedicated cost share to semaforo - For 2017	Gs.	(28,024,832)
Total Cost	Gs.	(5,662,922,298)
Total Funding	Gs.	198,000,000
Estimated Net Cost (Minus Fundings, CAFBANK for competitions) - 2017	Gs.	(5,464,922,298)

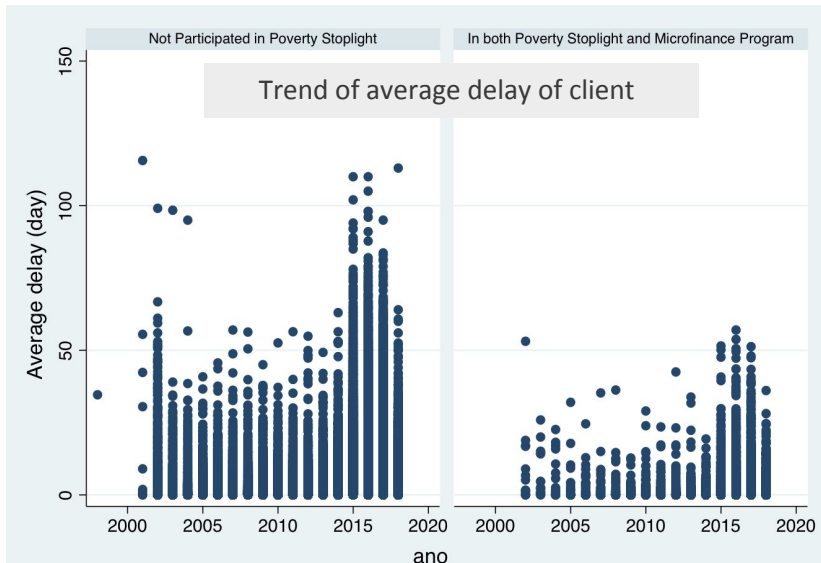
Has been Delivered to Nancy & Kathi

- First need to decide about 2 multipliers

- Dedication of asesora's time to only the semaforo - (First Number is based on the asesora questionnaire's result, in this questionnaire we ask from 30 asesoras about the their detail daily timing)
- Dedication of Headquarter cost for the semaforo excluding replica - (First Number is the ratio of "sum of the salaries of semaforo team excluding replicas/total HQ salaries)

Best scenario: ~25% time of each asesora is dedicated only to the semaforo, which can be even higher than that because of Research for the solutions & Transportation seems can be more time consuming.

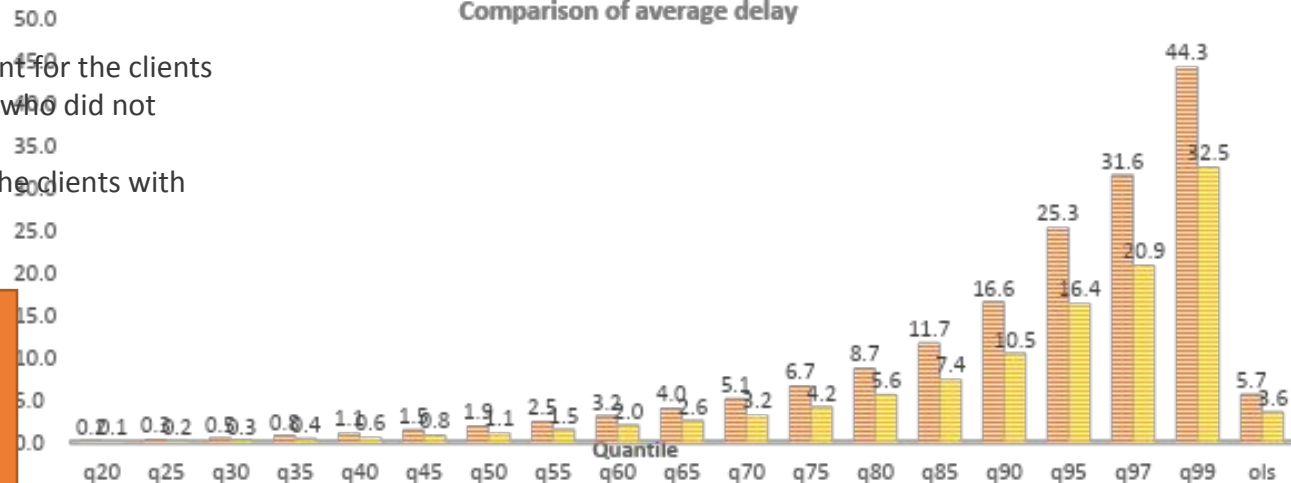
IMPACT ANALYSIS: Do PS's clients have lower repayment delay ?



- At First look we saw huge difference among the average repayment delay between the clients who are in the PS vs who are not
 - Suggested reason
 - PS impact only
 - MF committee impact
 - Self selection bias: to be qualified for PS client must not have more than 7 days of delay before the baseline survey

- And the average delay of repayment for the clients who participated in PS is less than who did not (controlling M.F. committee).
- Also, the difference increases for the clients with higher delays (higher quantiles)

Comparison of average delay



Big concern:
Self selection bias!

=> Maybe this results is due to selection of good behavior customers?

IMPACT ANALYSIS: Do PS's clients have lower repayment delay ?

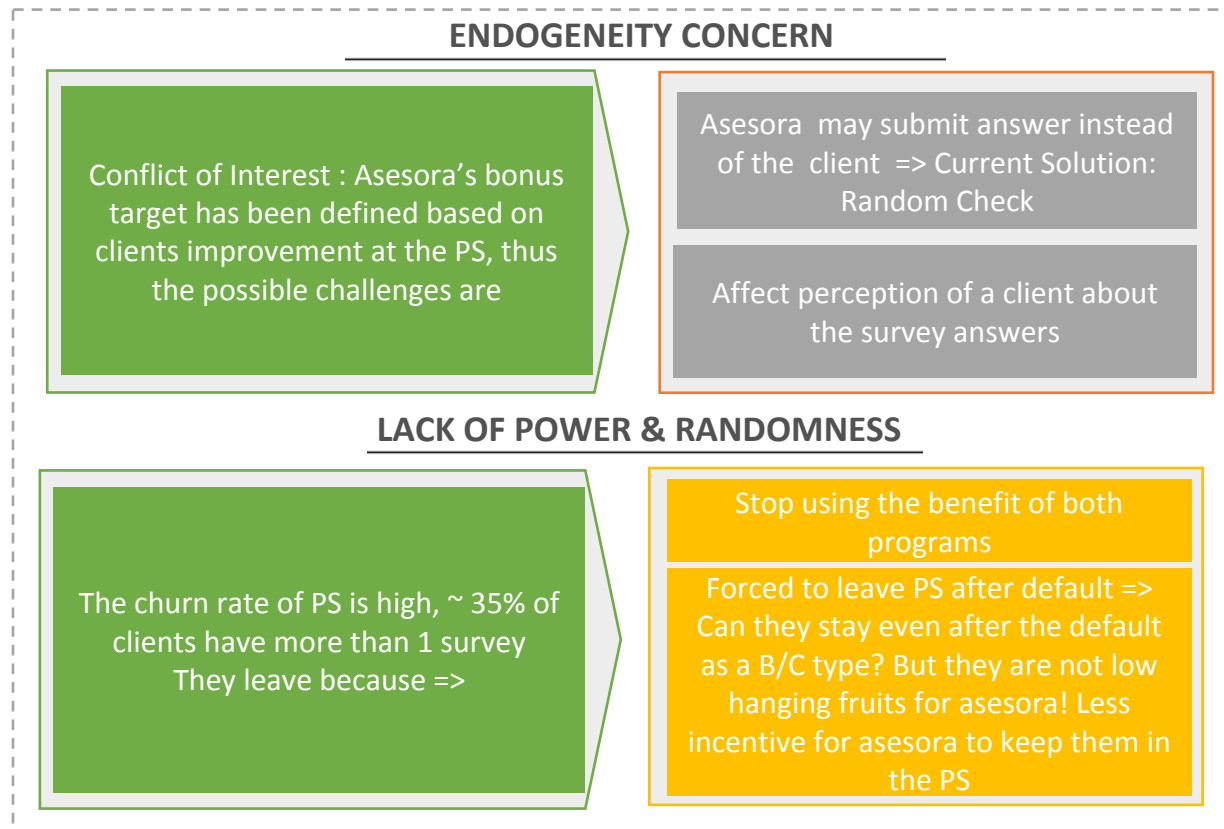
We use synthetic control group method (PSM) to create a more comparable control groups => mitigate the effect self-selection bias

VARIABLES	delay_day
_treated	-1.459***
	[0.122]
Constant	4.906***
	[0.0290]
Observations	69,323
R-squared	0.002
Standard errors in brackets	
With the assumption of common support	
*** p<0.01, ** p<0.05, * p<0.1	

And the average delay of repayment for the clients who participated in PS is 1.5 day less than who did not (Still controlling M.F. committee effect)

The **real impact can be higher** than this, because this is just the impact of PS on a sub group of the clients not whole population

Recommendations



Recommendations

Incentive Adjustment for the asesora

Future investigation for Higher impact to answer the cost effectiveness

Improve the churn rate of PS for future analysis

Deliverables to the PS team

Item	Description	Type
1 PS - Business Impact Quantitative Analysis - CafePar	The Model Report for quantitative analysis of empresas data	Model Report, excel file
2 PS - LitRev WorkCondition and Productivity	Short Literature review about the relation of work conditions on productivity, also the logic about the selected indicators in the the Model Report for Empresas	Report, word file
3 CafePar Databases		excel file
4 PS - Microfinance Data Analysis	Evaluation of the impact of the poverty stoplight on the microfinance clients; Do the PS clients have lower delay in repayment?	Report , word file
5 Analysis Do files		
6 Meta Data	Description of the variables in related databases	word file
7 PS - Semaforo Cost Report	Calculation of semaforo cost, this file also contains the result of the asesora questionnaire of timing	Model Report, excel file
8 PS - Final presentation	Final Presentation which contains the result of above documents	ppt file



Here we saw an amazing motivated team...



Muchas Gracias por todos los buenos momentos!



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Asesora's time dedications

Client's Group Type	Program Activities	Average required time (minutes) per month	Total number per month	Total Time for asesora (minuts)	Days
Group A	1 mandatory monthly visit - scheduling and meeting duration	40.3	28	1127.3	2.2
	1 mandatory monthly visit - transportation	49.0	11.2	548.4	1.1
	1 committee meeting (during Microfinance's meeting)	33.2	4	132.9	0.3
	1 replied contact	15.3	28	427.2	0.8
	Extra: helping clients to accomplish their goals, such as; bring them to dentist & ...	24.0	28	671.0	1.3
Group B	If necessary, 1 monthly visit - meeting duration	32.1	8.7	278.7	0.5
	If necessary, 1 monthly visit - transportation	49.0	3.5	170.2	0.3
	If necessary, 1 replied contact	15.3	13	198.4	0.4
	Extra: helping clients to accomplish their goals (Considered zero for group B)	0.0	7	0.0	0.0
Research	Training			120.0	0.2
Research	Research for PS solutions			1049.3	2.1
			Sum	4723.3	9.3
			Hours	78.7	
			Total number of asesora		
General Manager's Monthly Checking S	On Average 1 check per asesora	40.3	70	2818.1	
			hours per month, For one G. manager	0.7	
	Total hours for semaforo	78.7			
	Total working hours per month	187			
	Dedication Share	42%			
	Type A cost share <= calculated from the time dedications	82%			
	Type B cost share	18%			